

AN INTRODUCTION TO

PROBABILISTIC GRAPHICAL MODELS



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1 Graphical models and local dependence

Let $G = (\mathcal{V}, \mathcal{E})$ be a finite undirected graph. Each vertex $i \in \mathcal{V}$ indexes a random variable X_i , and an edge $\{i, j\} \in \mathcal{E}$ permits a direct interaction between X_i and X_j . The graph is not itself a probability distribution. It records which conditional dependences a distribution is allowed to have.

Definition 1.1 (Neighborhood). *The neighborhood of a vertex i is*

$$\mathcal{N}(i) = \{j \in \mathcal{V} : \{i, j\} \in \mathcal{E}\}. \quad (1.1)$$

Thus $\mathcal{N}(i)$ is the set of vertices joined directly to i by an edge.

Example 1.1 (The neighborhood of an interior grid vertex). In the 5×5 grid in Figure 1, the red vertex is i . Its four blue vertices are precisely $\mathcal{N}(i)$. The remaining white vertices are not neighbors of i .

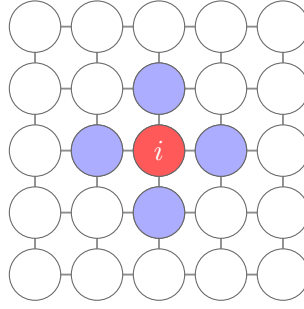


Figure 1: The red node is i ; the blue nodes form its neighborhood $\mathcal{N}(i)$.

Definition 1.2 (Separation). *Let $A, B, S \subseteq \mathcal{V}$ be pairwise disjoint. The set S separates A and B if every path from a vertex in A to a vertex in B contains at least one vertex of S .*

Example 1.2 (Separation in a four-cycle). Consider the cycle $A - B - C - D - A$. Every path from A to C passes through either B or D , so the set $\{B, D\}$ separates A and C . Likewise, every path from B to D passes through either A or C , so $\{A, C\}$ separates B and D . Figure 2 marks the separating vertices in blue.

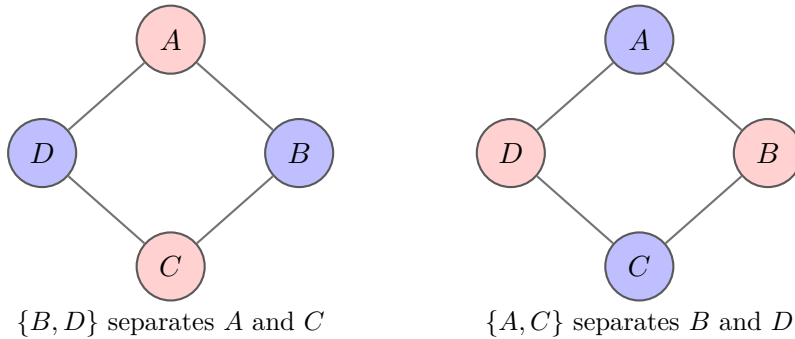


Figure 2: A separator blocks every graph path between the two red endpoint sets.

Definition 1.3 (Markov random field). *A joint distribution of $X = (X_i)_{i \in \mathcal{V}}$ is a Markov random field (MRF) with respect to G if, whenever S separates A and B ,*

$$X_A \perp X_B \mid X_S. \quad (1.2)$$

This implication is the global Markov property.

Remark 1.4. *Conditional independence in (1.2) says that, for values having positive conditioning probability,*

$$p(x_A \mid x_B, x_S) = p(x_A \mid x_S). \quad (1.3)$$

Equivalently, the conditional law of X_A given (X_B, X_S) is the same as its conditional law given X_S . Once X_S is known, observing X_B provides no further information about X_A .

Lemma 1.5 (A neighborhood separates a node from its non-neighbors). *For every vertex i , the set $\mathcal{N}(i)$ separates $\{i\}$ from*

$$R_i = \mathcal{V} \setminus (\{i\} \cup \mathcal{N}(i)).$$

Proof. Take any path from i to a vertex in R_i . Its first edge leaves i and therefore reaches a vertex adjacent to i . That vertex belongs to $\mathcal{N}(i)$. Hence every such path meets $\mathcal{N}(i)$. $\square$

Corollary 1.6 (The local Markov property). *If X is an MRF with respect to G , then*

$$X_i \perp X_{R_i} \mid X_{\mathcal{N}(i)} \quad (1.4)$$

and therefore

$$p(x_i \mid x_{-i}) = p(x_i \mid x_{\mathcal{N}(i)}). \quad (1.5)$$

Thus the conditional distribution of X_i depends only on neighboring variables.

The neighborhood $\mathcal{N}(i)$ is therefore a *Markov blanket* for node i : after its values are known, every non-neighbor can be ignored when computing the conditional distribution of X_i . This is the first computational gain supplied by the graph.

2 Hammersley–Clifford and clique factorization

We now fix a graph G and ask which positive distributions are MRFs with respect to it.

Definition 2.1 (Clique and maximal clique). *A set $C \subseteq \mathcal{V}$ is a clique if every pair of distinct vertices in C is joined by an edge. It is maximal if it is not properly contained in a larger clique.*

The following theorem gives the required characterization. Its strict positivity assumption is essential; see Lauritzen for a statement and proof [1].

Theorem 2.2 (Hammersley–Clifford). *Let $G = (\mathcal{V}, \mathcal{E})$ be a finite undirected graph, let every X_i have a finite state space, and suppose $p(x) > 0$ for every configuration x . Then p is an MRF with respect to G if and only if there are positive functions ψ_C on the maximal cliques of G such that*

$$p(x) = \frac{1}{Z} \prod_{C \in \mathcal{C}_{\max}} \psi_C(x_C), \quad Z = \sum_x \prod_{C \in \mathcal{C}_{\max}} \psi_C(x_C). \quad (2.1)$$

The functions ψ_C are called clique potentials. They need not themselves be probability distributions. The single constant Z normalizes their product.

Example 2.1 (The chain $A - B - C$). The graph in Figure 3 has maximal cliques $\{A, B\}$ and $\{B, C\}$.

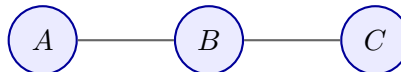


Figure 3: The three-variable chain $A - B - C$.

Hammersley–Clifford gives

$$p(a, b, c) = \frac{1}{Z} \psi_{AB}(a, b) \psi_{BC}(b, c). \quad (2.2)$$

The factorization displays the required conditional independence. For fixed b ,

$$\begin{aligned} p(a, c \mid b) &= \frac{\psi_{AB}(a, b) \psi_{BC}(b, c)}{\sum_{a', c'} \psi_{AB}(a', b) \psi_{BC}(b, c')} \\ &= \frac{\psi_{AB}(a, b)}{\sum_{a'} \psi_{AB}(a', b)} \frac{\psi_{BC}(b, c)}{\sum_{c'} \psi_{BC}(b, c')} \\ &= p(a \mid b) p(c \mid b). \end{aligned}$$

Hence $A \perp C \mid B$, exactly as graph separation predicts.

Example 2.2 (A rectangular cycle of Bernoulli variables). Let $A, B, C, D \in \{0, 1\}$ occupy the four vertices of the cycle in Figure 4. Its maximal cliques are its four edges, so a positive MRF has the form

$$p(a, b, c, d) = \frac{1}{Z} \psi_{AB}(a, b) \psi_{BC}(b, c) \psi_{CD}(c, d) \psi_{DA}(d, a). \quad (2.3)$$

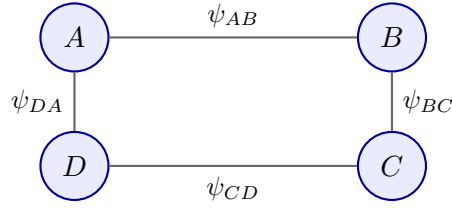


Figure 4: A four-node Bernoulli MRF with one potential on each maximal clique.

Conditioning on $B = b$ and $D = d$ leaves

$$p(a, c \mid b, d) \propto \underbrace{\psi_{AB}(a, b) \psi_{DA}(d, a)}_{\text{function of } a} \underbrace{\psi_{BC}(b, c) \psi_{CD}(c, d)}_{\text{function of } c}.$$

The normalizing sum factors into a sum over a times a sum over c . Therefore

$$p(a, c \mid b, d) = p(a \mid b, d) p(c \mid b, d),$$

so $A \perp C \mid (B, D)$. The same calculation gives $B \perp D \mid (A, C)$. These are precisely the two separation statements visible in the rectangle.

3 From clique potentials to energy models

Definition 3.1 (Clique energy and total energy). *For a positive clique potential ψ_C , define its energy by*

$$E_C(x_C) = -\log \psi_C(x_C). \quad (3.1)$$

The total energy of a configuration is

$$E(x) = \sum_{C \in \mathcal{C}_{\max}} E_C(x_C). \quad (3.2)$$

Substituting $\psi_C = e^{-E_C}$ into (2.1) gives the Boltzmann form

$$p(x) = \frac{e^{-E(x)}}{Z}, \quad Z = \sum_x e^{-E(x)}. \quad (3.3)$$

Lower energy means larger probability. The partition function Z is global: although every term in E is local, Z sums over every joint configuration.

The remaining modeling decision is the form of each function E_C . For continuous variables, a Taylor expansion motivates a local polynomial approximation. Keeping terms through degree two gives an affine-quadratic model. For binary variables $x_i \in \{0, 1\}$, polynomial expansions have a stronger interpretation: every function on a finite binary cube has an exact multilinear representation,

$$E(x) = c - \sum_i b_i x_i - \sum_{i < j} J_{ij} x_i x_j - \sum_{i < j < k} K_{ijk} x_i x_j x_k - \dots \quad (3.4)$$

Thus a quadratic binary energy is a genuine pairwise model, not merely a local approximation.

Quadratic truncation means that the energy can be factored using only vertex and edge terms. It does *not* force the graph itself to be triangle-free: a triangle can carry three pairwise edge interactions without a three-variable interaction. If the graph is triangle-free, however, its maximal cliques are exactly its vertices and edges. The two graphs used below—a rectangular grid and a bipartite graph—have this useful property.

Example 3.1 (A grid gives the Ising model). Place a spin $s_{r,c} \in \{-1, +1\}$ at every vertex of an $n \times n$ grid, and join horizontally or vertically adjacent sites. The graph is triangle-free. Its pairwise energy is

$$E(s) = - \sum_{\{u,v\} \in \mathcal{E}} J_{uv} s_u s_v - \sum_{u \in \mathcal{V}} h_u s_u. \quad (3.5)$$

This is the Ising model. A positive J_{uv} rewards equal neighboring spins because $s_u s_v = 1$ when they agree and -1 when they disagree. The field h_u controls the local preference at vertex u .

Figure 5 shows a representative finite grid and the two kinds of terms in (3.5).

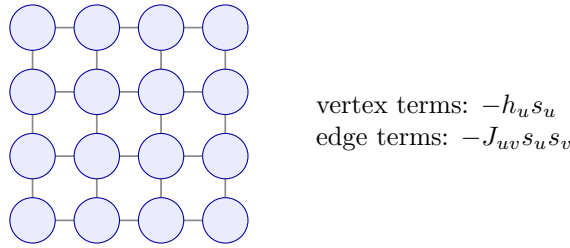


Figure 5: A finite rectangular Ising graph with vertex and nearest-neighbor edge energies.

When the vertices represent image pixels, neighboring pixels are encouraged to agree while observation-dependent vertex terms encourage fidelity to measured data. This is the basic MRF construction used in classical image restoration [2].

Example 3.2 (A bipartite graph gives a Bernoulli RBM). Split the vertices into m visible units $v_1, \dots, v_m$ and n hidden units $h_1, \dots, h_n$. A bipartite graph has edges only between the two groups, as illustrated in Figure 6. It cannot contain a triangle: traversing each edge switches groups, so after three edges one cannot return to the starting group.

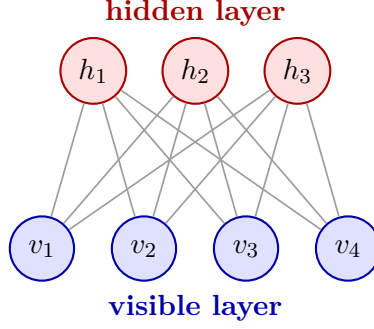


Figure 6: The bipartite graph of an RBM: visible–hidden edges are permitted, while edges within a layer are absent.

For Bernoulli states $v_i, h_j \in \{0, 1\}$, the restricted Boltzmann machine (RBM) has energy [4, 5]

$$E(v, h) = -a^\top v - b^\top h - v^\top W h = -\sum_i a_i v_i - \sum_j b_j h_j - \sum_{i,j} W_{ij} v_i h_j. \quad (3.6)$$

The matrix $W \in \mathbb{R}^{m \times n}$ contains visible–hidden edge parameters. The vectors $a \in \mathbb{R}^m$ and $b \in \mathbb{R}^n$ contain one-vertex biases. There are no $v_i v_{i'}$ or $h_j h_{j'}$ terms because there are no edges within either layer. This restriction is what makes both layer conditionals factorize and allows all units in one layer to be sampled in parallel.

Example 3.3 (A Gaussian–Bernoulli RBM). If the observed variables are continuous, take $v_i \in \mathbb{R}$ while retaining binary hidden variables $h_j \in \{0, 1\}$. One convenient parameterization is [5]

$$E(v, h) = \sum_i \frac{(v_i - a_i)^2}{2\sigma_i^2} - \sum_j b_j h_j - \sum_{i,j} \frac{W_{ij}}{\sigma_i^2} v_i h_j, \quad \sigma_i^2 > 0. \quad (3.7)$$

For fixed v , the terms involving h_j are

$$-h_j \left(b_j + \sum_i \frac{W_{ij} v_i}{\sigma_i^2} \right),$$

so

$$p(h_j = 1 \mid v) = \text{sigmoid} \left(b_j + \sum_i \frac{W_{ij} v_i}{\sigma_i^2} \right). \quad (3.8)$$

For fixed h , completing the square in v_i gives

$$p(v_i \mid h) = \mathcal{N} \left(a_i + \sum_j W_{ij} h_j, \sigma_i^2 \right). \quad (3.9)$$

The graph has not changed; only the visible state space and its vertex energy have changed. A Bernoulli RBM is natural for binary data, while a Gaussian–Bernoulli RBM is a possible model for real-valued measurements after appropriate scaling.

4 Local conditional probabilities

The MRF property says that a node conditional depends only on its neighbors. For a binary pairwise energy, the dependence has an explicit logistic form.

Theorem 4.1 (The conditional sigmoid). *Let $G = (\mathcal{V}, \mathcal{E})$ be a finite undirected graph and let $X_i \in \{0, 1\}$ for every i . Suppose*

$$p(x) = \frac{e^{-E(x)}}{Z}, \quad E(x) = c - \sum_{i \in \mathcal{V}} b_i x_i - \sum_{\{i,j\} \in \mathcal{E}} J_{ij} x_i x_j, \quad (4.1)$$

where $J_{ij} = J_{ji}$. Then, for every vertex i and every configuration of the other variables,

$$p(X_i = 1 \mid X_{-i} = x_{-i}) = \text{sigmoid} \left(b_i + \sum_{j \in \mathcal{N}(i)} J_{ij} x_j \right). \quad (4.2)$$

Proof. Hold x_{-i} fixed. Every term not containing x_i is the same for $x_i = 0$ and $x_i = 1$; collect these terms in $K(x_{-i})$. Then

$$\begin{aligned} E(0, x_{-i}) &= K(x_{-i}), \\ E(1, x_{-i}) &= K(x_{-i}) - b_i - \sum_{j \in \mathcal{N}(i)} J_{ij} x_j. \end{aligned}$$

Write $r_i = b_i + \sum_{j \in \mathcal{N}(i)} J_{ij} x_j$. Normalizing the two possible states gives

$$\begin{aligned} p(X_i = 1 \mid x_{-i}) &= \frac{e^{-E(1, x_{-i})}}{e^{-E(0, x_{-i})} + e^{-E(1, x_{-i})}} \\ &= \frac{e^{r_i}}{1 + e^{r_i}} = \text{sigmoid}(r_i). \end{aligned}$$

Only neighbors occur in r_i , as required by the Markov blanket property. $\square$

Corollary 4.2 (RBM layer conditionals). *For the Bernoulli RBM energy (3.6),*

$$p(h_j = 1 \mid v) = \text{sigmoid} \left(b_j + \sum_i W_{ij} v_i \right), \quad (4.3)$$

$$p(v_i = 1 \mid h) = \text{sigmoid} \left(a_i + \sum_j W_{ij} h_j \right). \quad (4.4)$$

Moreover,

$$p(h \mid v) = \prod_j p(h_j \mid v), \quad p(v \mid h) = \prod_i p(v_i \mid h). \quad (4.5)$$

Proof. Each hidden unit is adjacent only to visible units, so Theorem 4.1 gives (4.3); the visible calculation is identical. Once v is fixed, the RBM energy is a sum of separate functions of the individual h_j , so the conditional distribution factorizes over j . Fixing h gives the second factorization. $\square$

This is the central computational advantage of the RBM. The joint distribution couples the two layers, but conditioning on either layer makes all variables in the other layer independent.

5 MAP inference and ICM

Let y be an observed corrupted binary image and let x be the unknown clean image. An energy $E(x; y)$ defines

$$p(x \mid y) = \frac{e^{-E(x; y)}}{Z(y)}. \quad (5.1)$$

Because $Z(y)$ does not depend on x ,

$$\hat{x}_{\text{MAP}} = \arg \max_x p(x \mid y) = \arg \min_x E(x; y). \quad (5.2)$$

MAP inference is therefore energy minimization.

Iterated conditional modes (ICM), introduced by Besag for image restoration [3], is coordinate descent on the energy. It visits the variables in a fixed order and replaces each state by its lowest-energy local choice.

Algorithm 1 Iterated conditional modes (ICM)

Require: Energy $E(x; y)$, observed data y , initial configuration x , fixed ordering of the vertices

```

1: repeat
2:    $x^{\text{old}} \leftarrow x$ 
3:   for each vertex  $i$  in the fixed order do
4:      $x_i \leftarrow \arg \min_{z \in \mathcal{X}_i} E(z, x_{-i}; y)$ , retaining the current value if there is a tie
5:   end for
6: until  $x = x^{\text{old}}$ 
7: return  $x$ 
```

Only terms involving x_i are required for its update. Every actual change lowers the energy; on a finite state space, ICM must stop at a configuration that cannot be improved by changing one coordinate. This is a local minimum, not necessarily the global MAP solution.

Example 5.1 (Removing an isolated corrupted pixel on a rectangular grid). Let $x_i, y_i \in \{0, 1\}$ and let the graph be a rectangular pixel grid. Use

$$E(x; y) = \lambda \sum_i \mathbf{1}[x_i \neq y_i] + \gamma \sum_{\{i, j\} \in \mathcal{E}} \mathbf{1}[x_i \neq x_j], \quad \lambda, \gamma > 0. \quad (5.3)$$

The first term rewards fidelity to the observation; the second rewards agreement between neighboring clean pixels.

Consider the observed 3×3 patch in Figure 7, where white is 0 and black is 1. The center observation is 1 and its four grid neighbors are 0. Take $\lambda = 1$ and $\gamma = 1$.

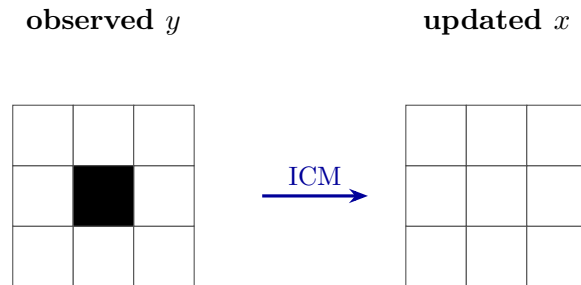


Figure 7: One ICM update removes an isolated black pixel when neighbor agreement outweighs fidelity to that observation.

If the center is kept at 1, its local energy is $0 + 4\gamma = 4$: it agrees with the observation but disagrees with four neighbors. If it is changed to 0, its local energy is $\lambda + 0 = 1$. ICM therefore flips the center to 0. The calculation uses only the center’s Markov blanket.

6 Gibbs sampling

6.1 Sampling

Suppose a random variable X can take values in a finite set $\mathcal{X}$ and has probability mass function p . To *sample from* p means to carry out a random procedure whose output is one value $x \in \mathcal{X}$ and satisfies

$$\Pr(\text{the procedure outputs } x) = p(x) \quad \text{for every } x \in \mathcal{X}. \quad (6.1)$$

The distribution p is the rule assigning probabilities. Running that rule once produces one value x , called a sample from p .

Example 6.1 (One Bernoulli sample). Let

$$p(X = 1) = 0.7, \quad p(X = 0) = 0.3.$$

Draw a number U uniformly from the interval $[0, 1]$ and return

$$X = \begin{cases} 1, & U \leq 0.7, \\ 0, & U > 0.7. \end{cases} \quad (6.2)$$

The interval assigned to 1 has length 0.7, so the procedure returns 1 with probability 0.7. The remaining interval has length 0.3, so it returns 0 with probability 0.3.

Figure 8 represents the two outcomes as intervals whose lengths equal their probabilities.



Figure 8: Sampling a Bernoulli variable by locating a uniform draw U in one of two probability intervals.

The following minimal Python program implements exactly the thresholding rule in (6.2).

Listing 1: One sample from $\text{Bernoulli}(0.7)$ using a uniform threshold.

```
import random

u = random.random() # Uniform(0, 1)
x = 1 if u <= 0.7 else 0
print(x)
```

Each execution returns one of the two possible states, 0 or 1. If the procedure is repeated independently many times, about 70% of the outputs will be 1. The notation $X \sim \text{Bernoulli}(0.7)$ describes the random rule; an observed value such as $x = 1$ is one sample from it.

The same construction works for more than two states: divide $[0, 1]$ into intervals whose lengths are the required probabilities and return the state whose interval contains U . Software libraries implement this operation directly, so in an algorithm the instruction

$$X \sim p$$

means “generate one random value according to the probabilities specified by p .”

Conditional sampling uses the current neighboring values to compute the probabilities for the next draw. For a concrete instance, suppose node i has two neighbors, $\mathcal{N}(i) = \{j, k\}$, and the pairwise model of Theorem 4.1 has

$$b_i = -\log 4, \quad J_{ij} = J_{ik} = \log 4.$$

If the current neighboring states are $x_j = x_k = 1$, the conditional logit is

$$b_i + J_{ij}x_j + J_{ik}x_k = -\log 4 + \log 4 + \log 4 = \log 4.$$

Equation (4.2) therefore gives

$$p(X_i = 1 \mid x_j = 1, x_k = 1) = \text{sigmoid}(\log 4) = \frac{4}{5} = 0.8.$$

This 0.8 is the threshold used for the draw: generate a fresh $U \sim \text{Uniform}(0, 1)$ and set $X_i = 1$ when $U \leq 0.8$, otherwise set $X_i = 0$.

The role of the neighbors is visible if one of them changes. If $x_j = 1$ and $x_k = 0$, the logit becomes

$$-\log 4 + \log 4 + 0 = 0,$$

so $p(X_i = 1 \mid x_j = 1, x_k = 0) = \text{sigmoid}(0) = 0.5$. The new neighboring configuration therefore changes the threshold for the next sample from 0.8 to 0.5.

6.2 Gibbs sampling

Definition 6.1 (One-site Gibbs sampler). *Let π be a joint probability distribution on a finite product space*

$$\mathcal{X} = \mathcal{X}_1 \times \cdots \times \mathcal{X}_d.$$

Starting from a current configuration $x^{(t)}$, choose a coordinate i_t and form the next configuration by

$$X_{i_t}^{(t+1)} \sim \pi(\cdot \mid x_{-i_t}^{(t)}), \tag{6.3}$$

$$X_r^{(t+1)} = x_r^{(t)} \quad (r \neq i_t). \tag{6.4}$$

Thus only coordinate i_t is redrawn; every other coordinate retains its current value. Repeating these updates in a fixed cyclic order $1, \dots, d, 1, \dots$ defines the one-site Gibbs sampler. The resulting sequence of complete configurations is a Markov chain.

We can now compare the two update rules using the conditional distribution calculated above. When $x_j = x_k = 1$, the probabilities for X_i are

$$\pi(X_i = 0 \mid x_j = 1, x_k = 1) = 0.2, \quad \pi(X_i = 1 \mid x_j = 1, x_k = 1) = 0.8.$$

The ICM update sets $x_i = 1$, the state with the lowest conditional energy. The Gibbs update in (6.3) instead redraws X_i : it returns 1 with probability 0.8 and 0 with probability 0.2. ICM is a deterministic energy-minimization update; the Gibbs sampler is a Markov chain built from conditional random draws.

For an RBM it is more efficient to update a whole layer at once. Given $v^{(t)}$, draw all hidden units independently from $p(h \mid v^{(t)})$; then draw all visible units independently from $p(v \mid h^{(t)})$. Figure 9 shows these alternating conditional draws.

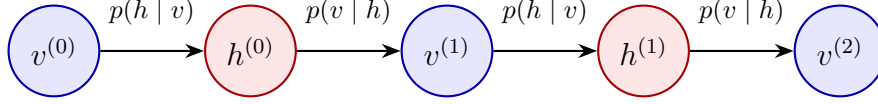


Figure 9: Alternating block-Gibbs updates in an RBM. Blue nodes are visible-layer states, red nodes are hidden-layer states, and each arrow denotes one conditional draw.

Proposition 6.2 (An RBM Gibbs sweep preserves the joint distribution). *Let (V, H) have the RBM joint distribution $p(v, h)$. Draw $V' \sim p(v | H)$ and then $H' \sim p(h | V')$. The resulting pair (V', H') also has distribution $p(v, h)$.*

Proof. After the visible redraw,

$$\Pr(V' = v', H = h) = \sum_v p(v, h) p(v' | h) = p(h) p(v' | h) = p(v', h).$$

After the hidden redraw,

$$\Pr(V' = v', H' = h') = \sum_h p(v', h) p(h' | v') = p(v') p(h' | v') = p(v', h').$$

Thus one complete sweep leaves the target joint distribution unchanged. $\square$

We use the following standard finite-state Markov-chain result [7].

Theorem 6.3 (Finite-state Gibbs convergence). *If a finite-state, time-homogeneous Markov chain is irreducible and aperiodic and has stationary distribution π , then its distribution converges to π from every starting state.*

For a finite Bernoulli RBM with finite parameters, every sigmoid probability lies strictly between 0 and 1. From any visible state, every hidden configuration therefore has positive conditional probability; from any hidden state, every visible configuration has positive conditional probability. A complete sweep can reach every joint configuration with positive probability, including the current configuration. The chain is consequently irreducible and aperiodic. The proposition identifies its stationary distribution, so repeated Gibbs sweeps converge to the RBM joint distribution.

7 Training an RBM

Let $v^{(1)}, \dots, v^{(N)}$ be observed training vectors. Hidden variables are not observed, so the visible marginal is

$$p_\theta(v) = \sum_h p_\theta(v, h) = \frac{\sum_h e^{-E_\theta(v, h)}}{Z_\theta}. \quad (7.1)$$

The maximum-likelihood objective is the average log-likelihood

$$\ell(\theta) = \frac{1}{N} \sum_{n=1}^N \log p_\theta(v^{(n)}). \quad (7.2)$$

Theorem 7.1 (Positive phase minus negative phase). *For an energy model $p_\theta(v, h) = Z_\theta^{-1} e^{-E_\theta(v, h)}$,*

$$\nabla_\theta \log p_\theta(v) = -\mathbb{E}_{p_\theta(h|v)}[\nabla_\theta E_\theta(v, h)] + \mathbb{E}_{p_\theta(v, h)}[\nabla_\theta E_\theta(v, h)]. \quad (7.3)$$

Proof. From (7.1),

$$\log p_\theta(v) = \log \sum_h e^{-E_\theta(v,h)} - \log Z_\theta.$$

Differentiating the first term gives

$$-\frac{\sum_h e^{-E_\theta(v,h)} \nabla_\theta E_\theta(v,h)}{\sum_h e^{-E_\theta(v,h)}} = -\mathbb{E}_{p_\theta(h|v)}[\nabla_\theta E_\theta(v,h)].$$

Likewise,

$$\nabla_\theta \log Z_\theta = -\mathbb{E}_{p_\theta(v,h)}[\nabla_\theta E_\theta(v,h)].$$

Subtracting this derivative proves (7.3). $\square$

For the Bernoulli RBM,

$$-\frac{\partial E}{\partial W_{ij}} = v_i h_j, \quad -\frac{\partial E}{\partial a_i} = v_i, \quad -\frac{\partial E}{\partial b_j} = h_j.$$

Consequently,

$$\frac{\partial \ell}{\partial W_{ij}} = \mathbb{E}_{p_{\text{data}}(v)p_\theta(h|v)}[V_i H_j] - \mathbb{E}_{p_\theta(v,h)}[V_i H_j], \quad (7.4)$$

$$\frac{\partial \ell}{\partial a_i} = \mathbb{E}_{p_{\text{data}}}[V_i] - \mathbb{E}_{p_\theta}[V_i], \quad (7.5)$$

$$\frac{\partial \ell}{\partial b_j} = \mathbb{E}_{p_{\text{data}}(v)p_\theta(h|v)}[H_j] - \mathbb{E}_{p_\theta(v,h)}[H_j]. \quad (7.6)$$

The first expectation in each line is the *positive phase*: statistics under observed data and their hidden explanations. The second is the *negative phase*: statistics under the model itself. The positive phase is easy because $p(h | v)$ factorizes. The negative phase is difficult because exact sampling from the model requires equilibrium Gibbs samples.

Definition 7.2 (Contrastive divergence). *Initialize a Gibbs chain with $V^{(0)} \sim p_{\text{data}}$ and $H^{(0)} \sim p_\theta(h | V^{(0)})$. After k complete Gibbs sweeps, let its distribution be $q_{k,\theta}$. If*

$$S_\theta(v, h) = -\nabla_\theta E_\theta(v, h),$$

the CD- k direction is

$$g_k(\theta) = \mathbb{E}[S_\theta(V^{(0)}, H^{(0)})] - \mathbb{E}[S_\theta(V^{(k)}, H^{(k)})]. \quad (7.7)$$

For a batch of B independent chains, its Monte Carlo estimator is

$$\hat{g}_{k,B}(\theta) = \frac{1}{B} \sum_{n=1}^B [S_\theta(V_n^{(0)}, H_n^{(0)}) - S_\theta(V_n^{(k)}, H_n^{(k)})]. \quad (7.8)$$

This data-initialized replacement of the equilibrium negative phase is the CD- k construction introduced by Hinton [6].

Theorem 7.3 (What the CD estimator computes). *Fix the parameter θ and the chain length k .*

(i) The estimator is unbiased for the CD- k direction:

$$\mathbb{E}[\hat{g}_{k,B}(\theta)] = g_k(\theta).$$

(ii) As $B \rightarrow \infty$,

$$\hat{g}_{k,B}(\theta) \longrightarrow g_k(\theta) \quad \text{almost surely.}$$

(iii) For a finite Bernoulli RBM with finite parameters,

$$g_k(\theta) \longrightarrow \nabla_{\theta} \ell(\theta) \quad \text{as } k \rightarrow \infty.$$

Proof. The summands in (7.8) are independent and identically distributed, and their common expectation is exactly (7.7). This proves (i). Every Bernoulli RBM statistic is bounded, so the strong law of large numbers applied coordinatewise proves (ii).

For (iii), Section 6 established that the endpoint distribution $q_{k,\theta}$ converges to the RBM joint distribution $p_{\theta}(v, h)$. Since S_{θ} is bounded on the finite state space,

$$\mathbb{E}_{q_{k,\theta}}[S_{\theta}(V, H)] \longrightarrow \mathbb{E}_{p_{\theta}(v, h)}[S_{\theta}(V, H)].$$

The initial expectation in (7.7) is the exact positive phase. The limit of the endpoint expectation is the exact negative phase. The likelihood-gradient identity therefore gives the claimed limit. $\square$

The theorem gives a precise correctness statement at a fixed parameter value: a large batch accurately estimates the CD- k direction, and a sufficiently long Gibbs chain makes that direction approach the maximum-likelihood gradient. For finite k , CD is generally a biased approximation to the exact gradient [8, 9]. Convergence of the full parameter recursion is a separate question from the fixed-parameter sampling and Monte Carlo claims proved here.

Algorithm 2 One mini-batch CD- k training update for a Bernoulli RBM

Require: Batch $v_1^{(0)}, \dots, v_B^{(0)}$, parameters (W, a, b) , chain length k , learning rate η

1: Sample $h_n^{(0)} \sim p_{\theta}(h \mid v_n^{(0)})$ independently for $n = 1, \dots, B$

2: **for** $r = 0, \dots, k - 1$ **do**

3: Sample $v_n^{(r+1)} \sim p_{\theta}(v \mid h_n^{(r)})$ for every n

4: Sample $h_n^{(r+1)} \sim p_{\theta}(h \mid v_n^{(r+1)})$ for every n

5: **end for**

6: $\hat{G}_W \leftarrow B^{-1} \sum_n [v_n^{(0)}(h_n^{(0)})^{\top} - v_n^{(k)}(h_n^{(k)})^{\top}]$

7: $\hat{G}_a \leftarrow B^{-1} \sum_n (v_n^{(0)} - v_n^{(k)})$

8: $\hat{G}_b \leftarrow B^{-1} \sum_n (h_n^{(0)} - h_n^{(k)})$

9: $W \leftarrow W + \eta \hat{G}_W$, $a \leftarrow a + \eta \hat{G}_a$, $b \leftarrow b + \eta \hat{G}_b$

The positive hidden samples in Algorithm 2 may be replaced by their conditional means

$$\mathbb{E}[H_j \mid v] = p(H_j = 1 \mid v),$$

which are available from (4.3). This keeps the expected positive-phase statistic unchanged and removes unnecessary sampling noise from that term.

8 Using a trained RBM

After training, the parameters (W, a, b) define a joint distribution over visible data and hidden features. The same two conditional distributions used during training provide the main deployment operations.

Feature extraction. For an observed vector v , compute

$$z_j(v) = p(H_j = 1 \mid v) = \text{sigmoid} \left(b_j + \sum_i W_{ij} v_i \right). \quad (8.1)$$

The vector $z(v)$ is a learned representation of the observation. It can be passed to a classifier or visualized to inspect which hidden features respond to the input.

Reconstruction and missing values. Given a corrupted or incomplete visible vector, alternate $h \sim p(h \mid v)$ and $v \sim p(v \mid h)$. If some coordinates are known, clamp those observed coordinates after each visible update and redraw only the missing coordinates. A deterministic reconstruction can use the conditional probabilities in place of Bernoulli draws, but it should be interpreted as a conditional mean rather than a sample.

Generation. Choose an initial visible vector and run the block-Gibbs chain from Section 6. After the chain has approached the model distribution, retained visible states are approximate samples from the trained model. The hidden states supply the latent explanations used to generate them.

Comparing visible configurations. Summing the Bernoulli hidden units analytically gives

$$p_\theta(v) = \frac{e^{-F_\theta(v)}}{Z_\theta}, \quad (8.2)$$

where

$$F_\theta(v) = -a^\top v - \sum_j \log \left(1 + \exp \left[b_j + \sum_i W_{ij} v_i \right] \right). \quad (8.3)$$

Indeed,

$$\begin{aligned} \sum_h e^{-E(v,h)} &= e^{a^\top v} \prod_j \sum_{h_j \in \{0,1\}} e^{h_j(b_j + \sum_i W_{ij} v_i)} \\ &= e^{a^\top v} \prod_j \left(1 + e^{b_j + \sum_i W_{ij} v_i} \right) = e^{-F_\theta(v)}. \end{aligned}$$

Because Z_θ is the same for every v , lower free energy means larger model probability. Thus $F_\theta(v)$ can rank visible configurations without computing the partition function. It does not by itself give a normalized probability.

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